

Review of Last Lecture

3.1 Introduction

3.2 Fourier Series Analysis of Periodic Signals

3.3 Fourier Series of Typical Periodic Signals

3.2 Fourier Series Analysis of Periodic Signals



Fourier Series Expansion: **Periodic signals** can be expressed as a **weighted sum** of **sinusoidal signals with harmonic relationships**.

- **Dirichlet conditions** (**sufficient but unnecessary conditions** for the existence of Fourier series):
 - (1) Within one period, the signal is **continuous** or has only **a finite number of first-kind discontinuities** (the function has finite left and right limits at the discontinuity, e.g., $x(t)=|\sin(t)|/\sin(t)$ at $t=0$).
 - (2) Within one period, the signal has **a finite number of maximum and minimum values**.
 - (3) Within one period, the signal is **absolutely integrable**, i.e., $\int_0^T |f(t)| dt < \infty$.
- **Signals in engineering all satisfy Dirichlet conditions.**

3.2 Fourier Series Analysis of Periodic Signals



3.2.1 Fourier Series in Trigonometric Form

Let $f(t)$ be a periodic signal with period T_1 .

Its **fundamental angular frequency** is $\omega_1 = 2\pi f_1 = \frac{2\pi}{T_1}$.

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t) \quad (1)$$

DC component: **Amplitude of cosine component :** **Amplitude of sine component:**

$$a_0 = \frac{1}{T_1} \int_{t_0}^{t_0+T_1} f(t) dt \qquad a_n = \frac{2}{T_1} \int_{t_0}^{t_0+T_1} f(t) \cos n\omega_1 t dt \qquad b_n = \frac{2}{T_1} \int_{t_0}^{t_0+T_1} f(t) \sin n\omega_1 t dt$$

The integral limits in the above formulas are generally set as: $0 \sim T_1$ or $-\frac{T_1}{2} \sim \frac{T_1}{2}$

The Fourier series in trigonometric form can also be expressed as:

$$f(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_1 t + \varphi_n) \quad (2)$$

where $c_0 = a_0$ $c_n^2 = a_n^2 + b_n^2$ $\varphi_n = \arctan\left(-\frac{b_n}{a_n}\right)$

3.2 Fourier Series Analysis of Periodic Signals



3.2.2 Fourier Series in Exponential Form

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t} \quad (3)$$

where $F_n = \frac{1}{T_1} \int_{t_0}^{t_0+T_1} f(t) e^{-jn\omega_1 t} dt$ -----Complex amplitude

$$F_0 = a_0 = c_0$$

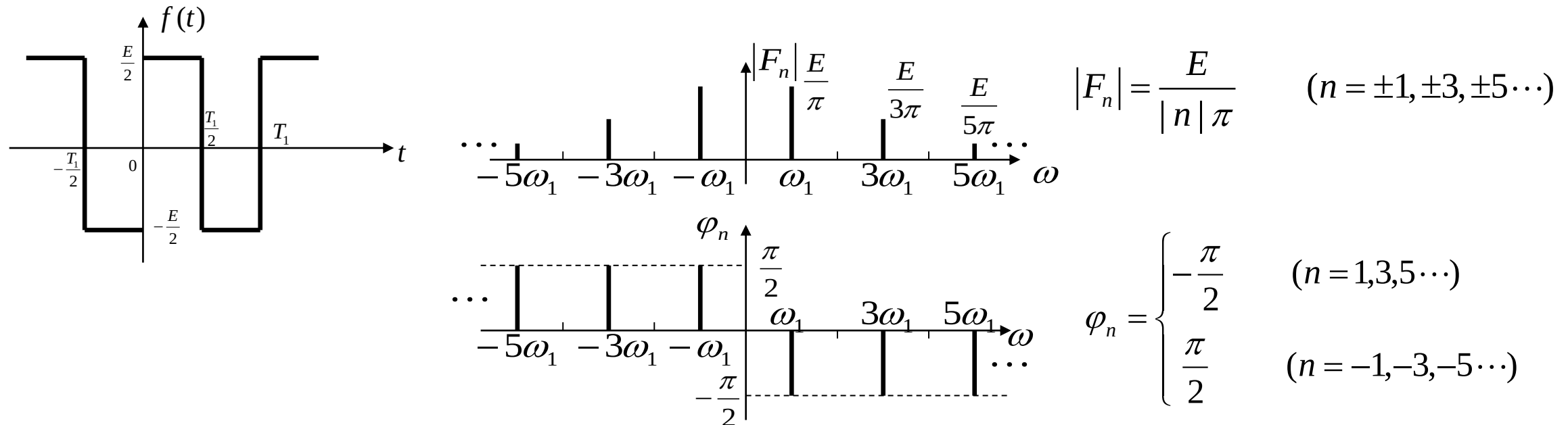
$$F_n = |F_n| e^{j\varphi_n} = \frac{1}{2} (a_n - jb_n)$$

$$|F_n| = \frac{1}{2} \sqrt{a_n^2 + b_n^2} = \frac{1}{2} c_n \quad \varphi_n = \arctan \left(-\frac{b_n}{a_n} \right)$$

3.2 Fourier Series Analysis of Periodic Signals

3.2.3 Characteristics of Periodic Signal Spectrum

- (1) **Discreteness:** The spectrum is discrete.
- (2) **Harmonicity:** Spectral lines appear at integer multiples of the fundamental frequency ω_1 .
- (3) **Convergence:** The amplitude of spectral lines in the amplitude spectrum gradually decays to zero as n increases ($n \rightarrow \infty$).



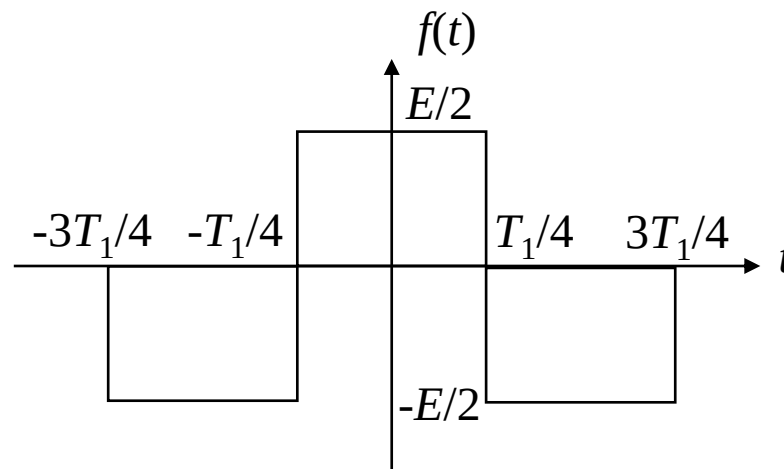
The zero components of the Fourier series in trigonometric form for the periodic rectangular signal shown in the figure are ().

☒ A $a_n = 0 \quad (n = 0, 2, 4 \dots)$

☐ B $a_n = 0 \quad (n = 1, 3, 5 \dots)$

☒ C $b_n = 0 \quad (n = 1, 3, 5 \dots)$

☒ D $b_n = 0 \quad (n = 2, 4, 6 \dots)$



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3.2 Fourier Series Analysis of Periodic Signals

3.2.4 Relationship Between Waveform Symmetry and Harmonic Characteristics

If $f(t)$ is a real function and its waveform satisfies a certain **symmetry**, some terms in the Fourier series disappear.

Waveform Symmetry	Frequency Components
Even Function $f(t) = f(-t)$	No sine components. $b_n = 0 \ (n = 1, 2, 3 \dots)$
Odd Function $f(t) = -f(-t)$	No DC and cosine components. $a_n = 0 \ (n = 0, 1, 2, 3 \dots)$
Odd Harmonic Function (Half-Wave Odd Symmetry) $f(t \pm \frac{T_1}{2}) = -f(t)$	No DC and even harmonic components. $a_n = 0 \ (n = 0, 2, 4, 6 \dots)$ $b_n = 0 \ (n = 2, 4, 6 \dots)$
Even Harmonic Function (Half-Wave Even Symmetry) $f(t \pm \frac{T_1}{2}) = f(t)$	No odd harmonic components. $a_n = 0 \ (n = 1, 3, 5 \dots)$ $b_n = 0 \ (n = 1, 3, 5 \dots)$

3.2 Fourier Series Analysis of Periodic Signals



Which components of the Fourier series in trigonometric form for the periodic rectangular signal shown in the figure are zero?

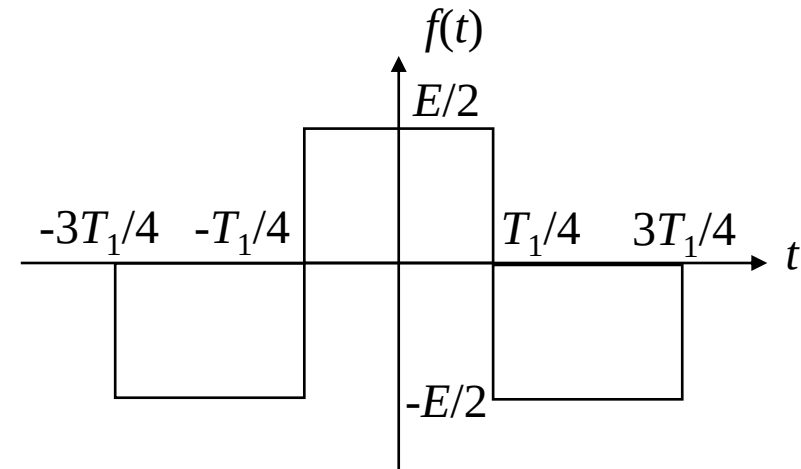
This is an **even function**, so the sine components are zero.

$$b_n = 0 \quad (n = 1, 2, 3, \dots)$$

It is also an **odd harmonic function** (half-wave odd symmetry), so the DC and even harmonic components are zero.

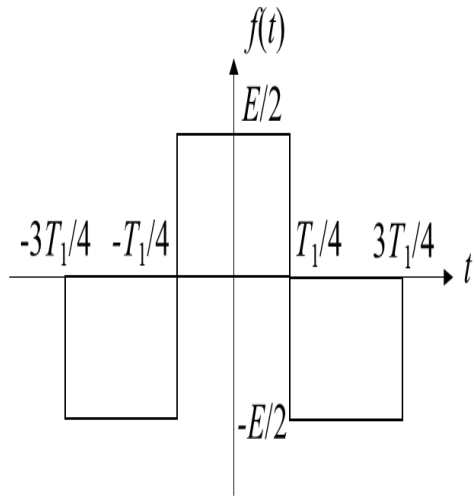
$$a_n = 0 \quad (n = 0, 2, 4, \dots)$$

$$b_n = 0 \quad (n = 2, 4, 6, \dots)$$



3.2 Fourier Series Analysis of Periodic Signals

Compared with the example in the last lecture, the two periodic rectangular signals have a time difference of $T_1/4$, and the corresponding trigonometric function phase difference is $\pi/2$.



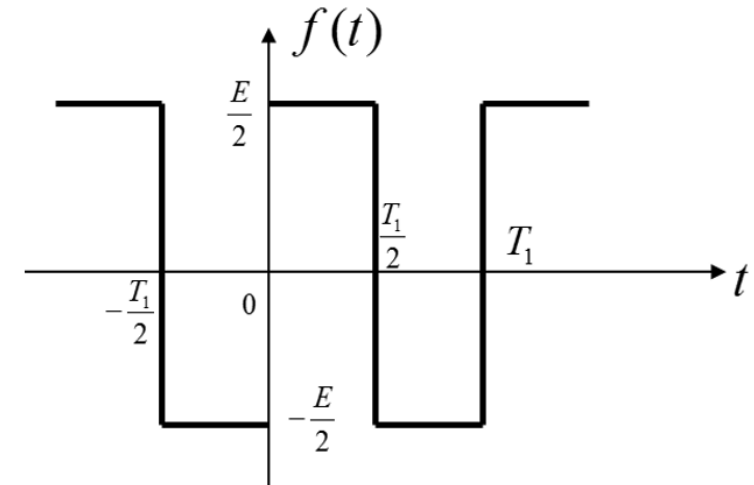
The zero components of the Fourier series in trigonometric form for the periodic rectangular signal shown in the figure are ().

A $a_n = 0 \quad (n = 0, 2, 4 \dots)$

B $a_n = 0 \quad (n = 1, 3, 5 \dots)$

C $b_n = 0 \quad (n = 1, 3, 5 \dots)$

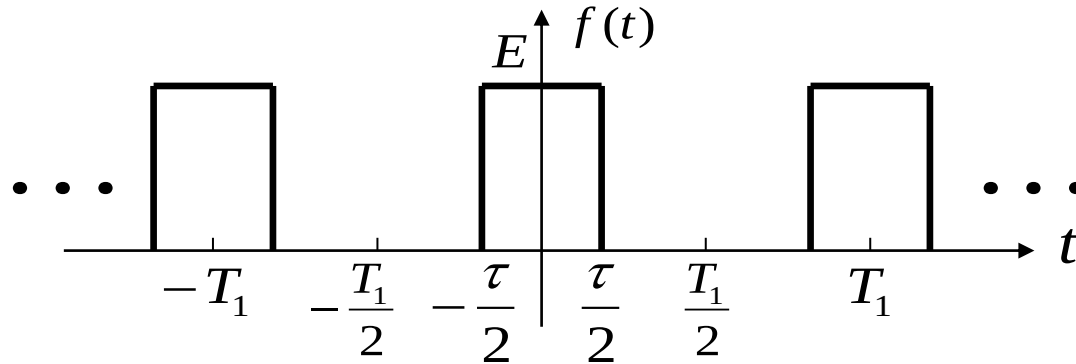
D $b_n = 0 \quad (n = 2, 4, 6 \dots)$



3.3 Fourier Series of Typical Periodic Signals



Fourier Series of Periodic Rectangular Pulse Signals



$b_n = 0$ (The Fourier series of an even function does not contain sine terms)

$$a_0 = \frac{2}{T_1} \int_0^{\frac{T_1}{2}} f(t) dt = \frac{E\tau}{T_1} \quad a_n = \frac{4}{T_1} \int_0^{\frac{T_1}{2}} f(t) \cos n\omega_1 t dt = \frac{2E\tau}{T_1} \text{Sa}\left(\frac{n\omega_1\tau}{2}\right) = c_n$$

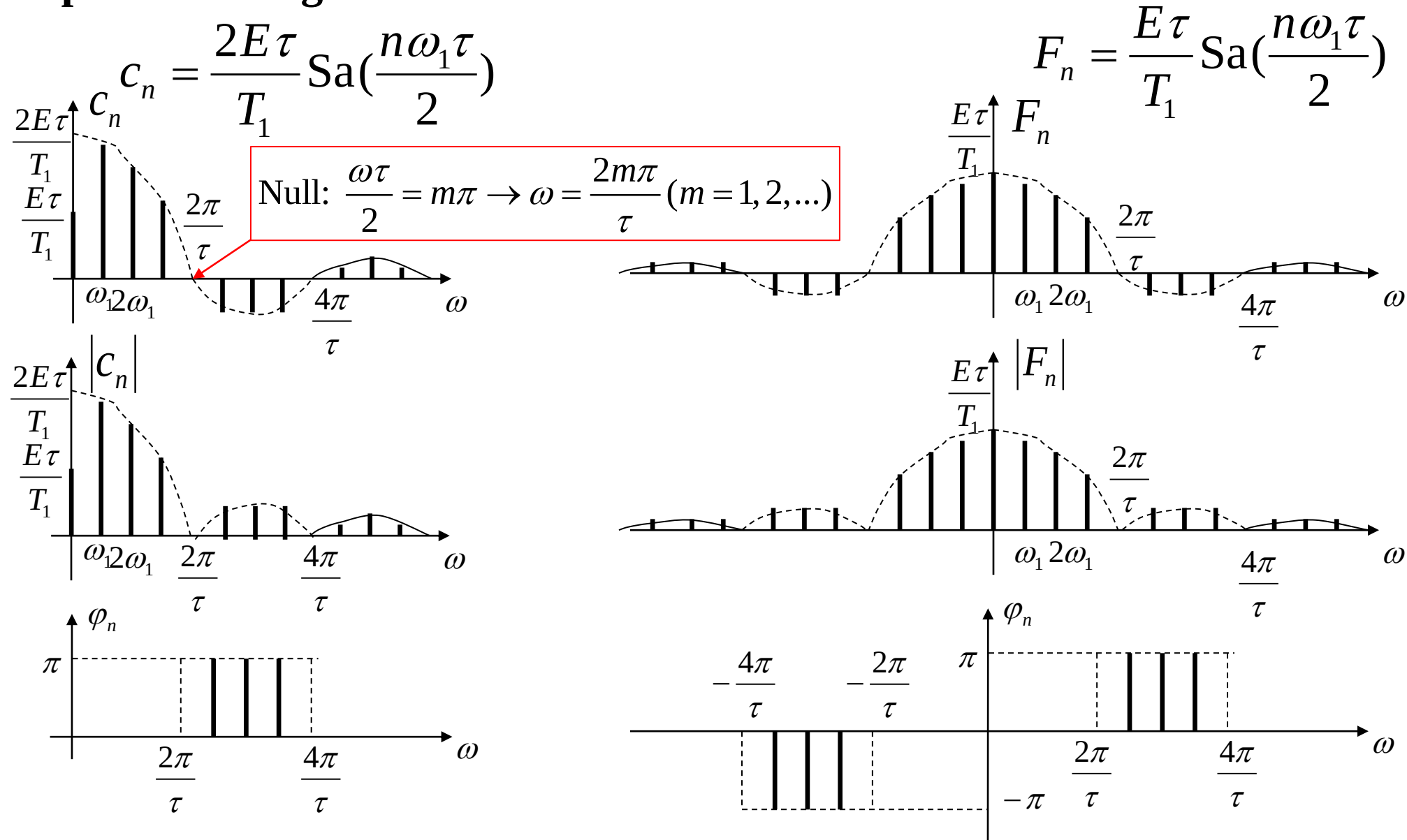
$$f(t) = \frac{E\tau}{T_1} + \frac{2E\tau}{T_1} \sum_{n=1}^{\infty} \text{Sa}\left(\frac{n\omega_1\tau}{2}\right) \cos n\omega_1 t$$

$$f(t) = \frac{E\tau}{T_1} \sum_{n=-\infty}^{\infty} \text{Sa}\left(\frac{n\omega_1\tau}{2}\right) e^{jn\omega_1 t} \quad F_n = \frac{1}{2}(a_n - jb_n) = \frac{1}{2}a_n = \frac{E\tau}{T_1} \text{Sa}\left(\frac{n\omega_1\tau}{2}\right)$$

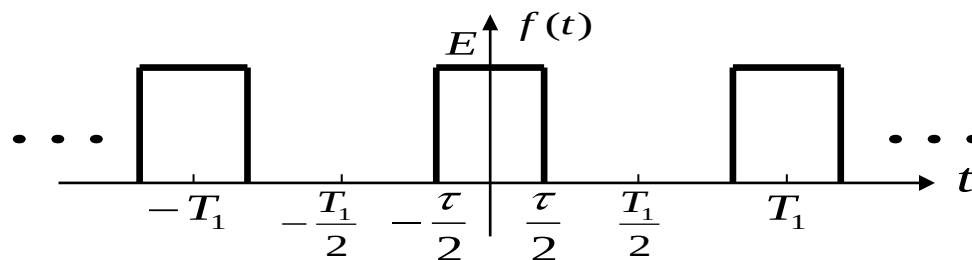
3.3 Fourier Series of Typical Periodic Signals



Spectrum Diagram



For the periodic rectangular signal waveform shown in the figure, which of the following statements are correct? ()



- ☒ A If T_1 is doubled, the spectral line spacing of the signal spectrum is halved, and the amplitude is halved.
- ☐ B If T_1 is doubled, the signal bandwidth is halved, and the amplitude is doubled.
- ☐ C If τ is halved, the spectral line spacing of the signal spectrum is halved, and the amplitude is doubled.
- ☒ D If τ is halved, the signal bandwidth is doubled, and the amplitude is halved.

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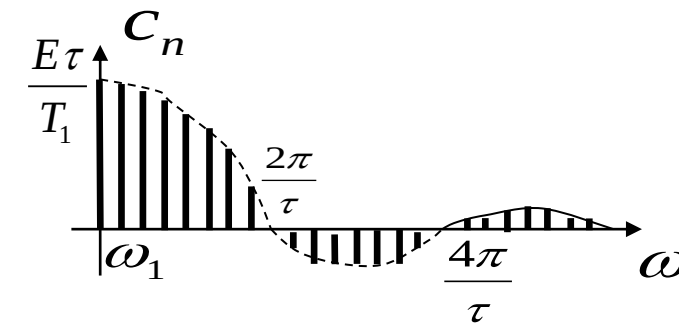
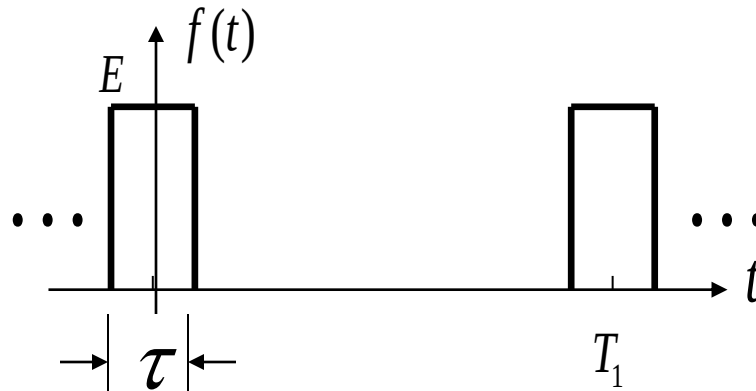
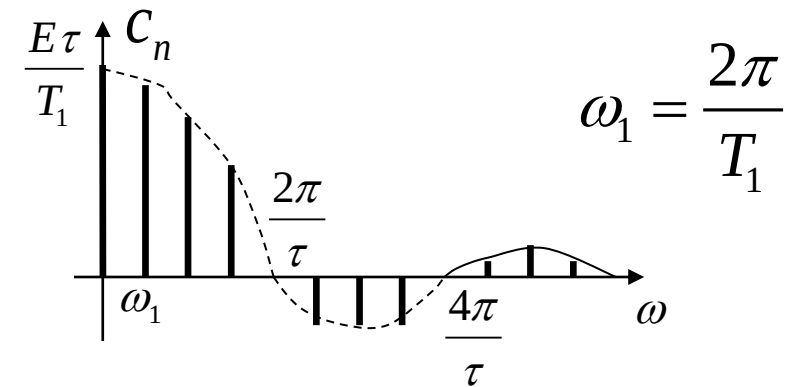
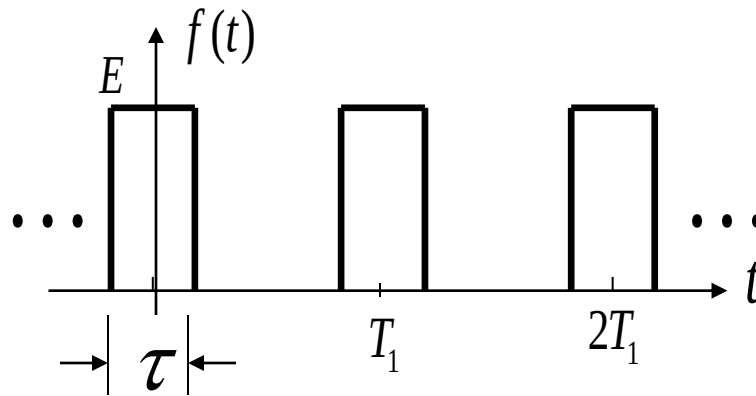
3.3 Fourier Series of Typical Periodic Signals



Relationship Between Spectrum Structure and Waveform Parameters (T_1, τ)

a) If τ remains unchanged and T_1 is doubled:

- The spectral line spacing of the amplitude spectrum is halved;
- The amplitude is halved.

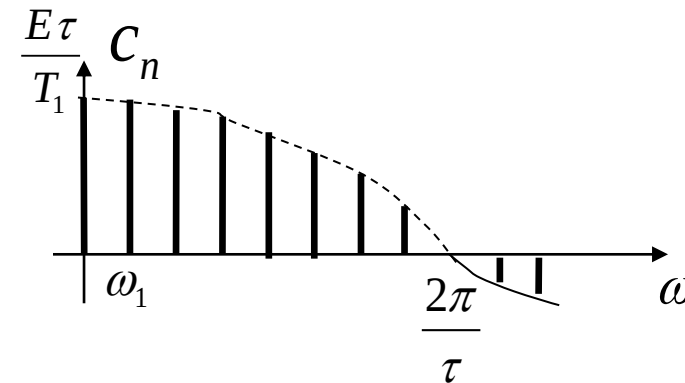
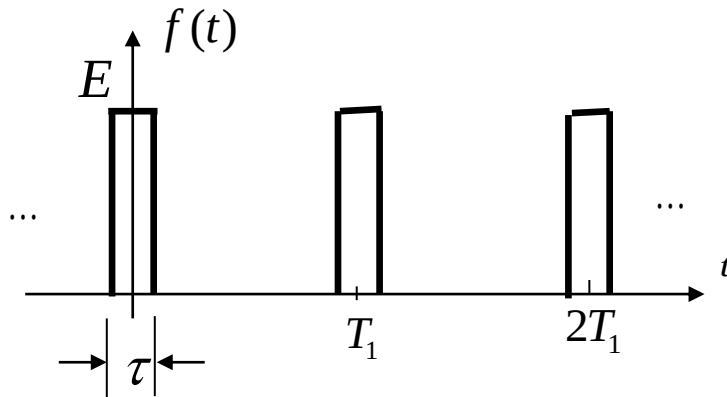
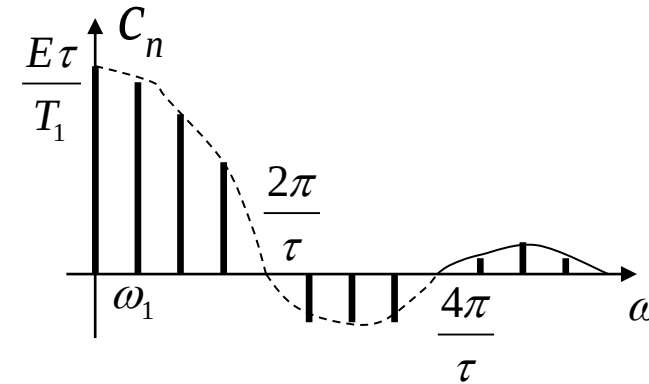
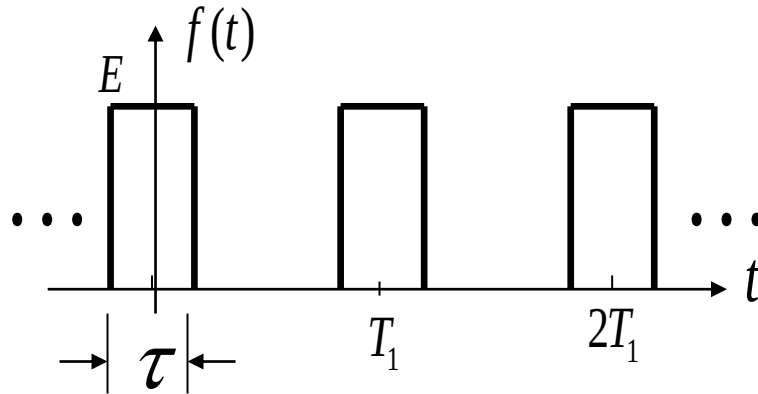


3.3 Fourier Series of Typical Periodic Signals



b) If T_1 remains unchanged and τ is halved:

- The signal bandwidth is doubled.
- The amplitude of the amplitude spectrum is halved.



Summary

- 1. Spectral line spacing** $\omega_1 (= \frac{2\pi}{T_1})$: Only related to T_1 , inversely proportional to T_1 .
- 2. Cross-zero frequency:** Only related to τ , inversely proportional to τ .
- 3. Spectral line amplitude:** Related to both T_1 and τ , inversely proportional to T_1 , directly proportional to τ .

Contents of This Lecture

3.4 Fourier Transform

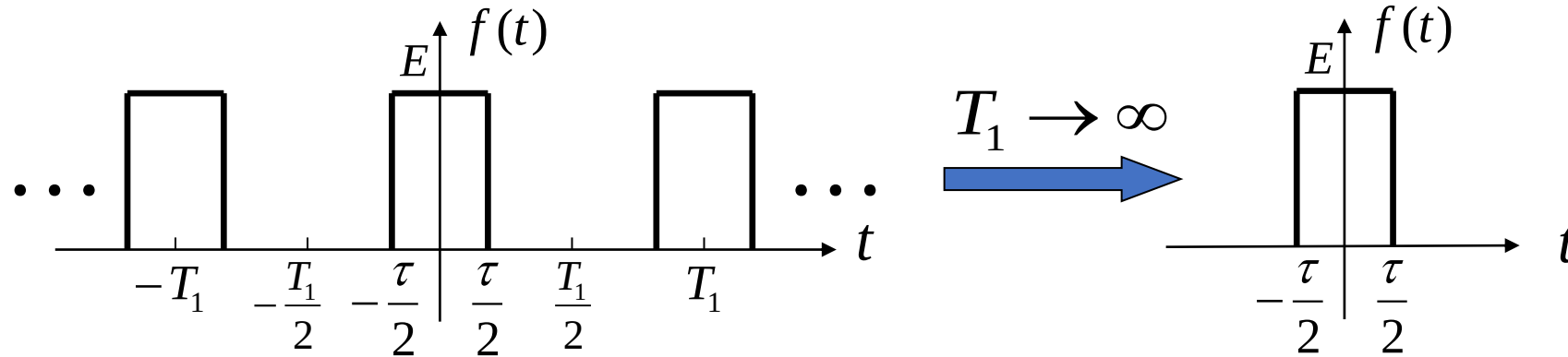
3.5 Fourier Transform of Typical Aperiodic Signals

3.6 Fourier Transform of Impulse and Step Functions

Learning Objectives of This Lecture

1. Gain an in-depth understanding of the connection and difference between **Fourier transform** and **Fourier series**.
2. Proficiently master the Fourier transform of **typical aperiodic signals, impulse functions, and step functions**.

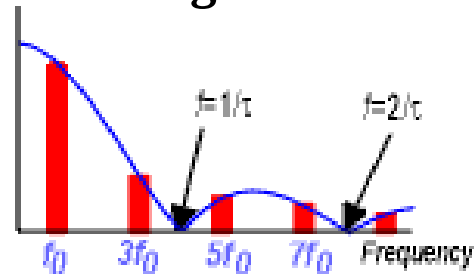
3.4 Fourier Transform



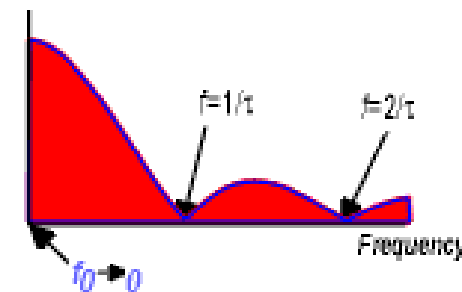
$T_1 \uparrow \longrightarrow \omega_1 = \frac{2\pi}{T_1} \downarrow \longrightarrow$ Spectral line spacing (proportional to ω_1) $\downarrow$

$T_1 \rightarrow \infty \longrightarrow \omega_1 = \frac{2\pi}{T_1} \rightarrow 0 \longrightarrow$ Spectral line spacing $\rightarrow 0$

Discrete Spectrum of Periodic Signals



Continuous Spectrum of Aperiodic Signals



3.4 Fourier Transform



Due to $T_1 \rightarrow \infty$, $F_n = \frac{1}{T_1} \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) e^{-jn\omega_1 t} dt \rightarrow 0$

Spectral Density Function: $\lim_{T_1 \rightarrow \infty} F_n T_1 = \lim_{T_1 \rightarrow \infty} \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) e^{-jn\omega_1 t} dt$

When $T_1 \rightarrow \infty$, discrete frequency $n\omega_1 \rightarrow$ continuous frequency ω .

Then $\lim_{T_1 \rightarrow \infty} F_n T_1 = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$

Let **$F(\omega)$** denote the **Fourier Transform of aperiodic signal $f(t)$** .

$$F(\omega) = \mathbf{F} [f(t)] = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

Inverse Fourier Transform

$$f(t) = \mathbf{F}^{-1} [F(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

$$F(\omega) = |F(\omega)| e^{j\varphi(\omega)}$$

$|F(\omega)|$: **Amplitude Spectrum**

$\varphi(\omega)$: **Phase Spectrum**

3.4 Fourier Transform



Physical Meaning of Fourier Transform

It can be proved,

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega = \int_0^{\infty} \underbrace{\frac{|F(\omega)|}{\pi}}_{\text{Amplitude}} d\omega \cos[\omega t + \varphi(\omega)] = \lim_{\Delta\omega \rightarrow 0} \sum_{n=0}^{\infty} \underbrace{\frac{|F(n\Delta\omega)|}{\pi}}_{\text{Amplitude}} \Delta\omega \cos[n\Delta\omega t + \varphi(n\Delta\omega)]$$

$f(t)$ is the sum of infinitely many continuous cosine signals with infinitesimal amplitude, $\left(\frac{1}{\pi}|F(\omega)|d\omega\right)$ covering frequencies from 0 to ∞ .

Aperiodic signals can be expressed as a **weighted integral of sinusoidal signals**, containing all **continuous frequency components from zero to infinitely high frequencies**.

For energy-limited signals (e.g., single-pulse signals), the amplitude of each frequency component, $\left(\frac{1}{\pi}|F(\omega)|d\omega\right)$ tends to be infinitely small. Thus, the spectrum can no longer be represented by amplitude, but by the **spectral density function** instead.

Sufficient but Not Necessary Condition for the Existence of Fourier Transform

$$\int_{-\infty}^{\infty} |f(t)| dt < \infty \quad \text{i.e., } f(t) \text{ is absolutely integrable.}$$

Proof: $\int_{-\infty}^{\infty} |f(t)e^{-j\omega t}| dt = \int_{-\infty}^{\infty} |f(t)| dt < \infty$

Most **energy signals** satisfy this condition.

If a signal does not satisfy the above condition, Fourier transform may still exist, and may be conducted by introducing the concept of **impulse function** $\delta(t)$.

Relationship Between Fourier Series & Transform

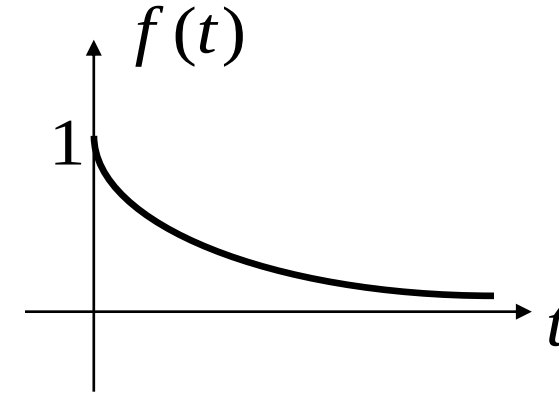
Feature	Fourier Series	Fourier Transform
Expression	$f(t) = \sum_{n=-\infty}^{+\infty} F_n e^{jn\omega_1 t}$	$f(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\omega) e^{j\omega t} d\omega$
Spectrum Type	Discrete Spectrum $F_n = \frac{1}{T_1} \int_{-T_1/2}^{T_1/2} f(t) e^{-jn\omega_1 t} dt$ $F_n = F_n e^{j\phi_n}$	Continuous Spectrum (Spectral Density) $F(\omega) = \int_{-\infty}^{+\infty} f(t) e^{-j\omega t} dt$ $F(\omega) = F(\omega) e^{j\phi(\omega)}$
Existence Condition	Dirichlet conditions within one period $\int_0^T f(t) dt < \infty$	Dirichlet conditions over $(-\infty, \infty)$ $\int_{-\infty}^{\infty} f(t) dt < \infty$

3.5 Fourier Transform of Typical Aperiodic Signals



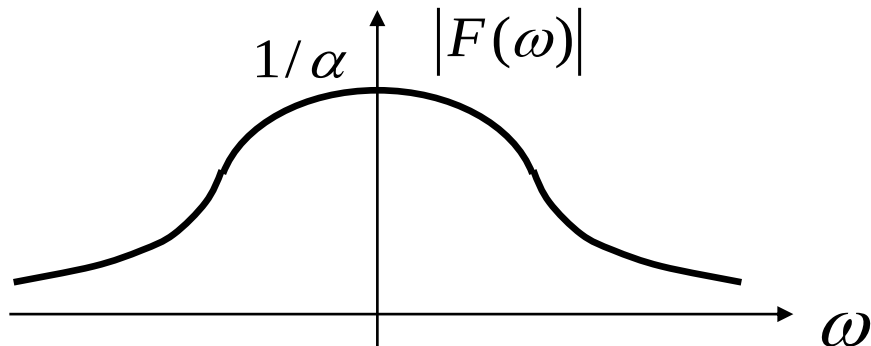
1. Single-sided Exponential Signal

$$f(t) = \begin{cases} e^{-\alpha t} & t > 0 \\ 0 & t < 0 \end{cases} = e^{-\alpha t} u(t)$$

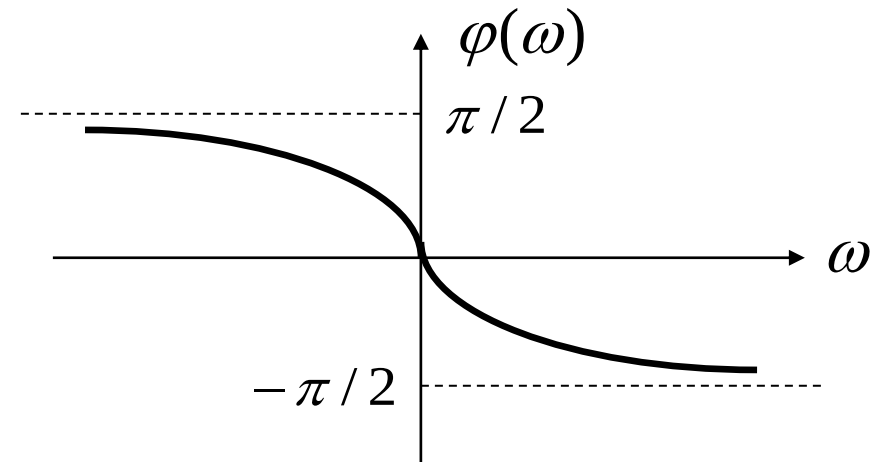


$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt = \int_0^{\infty} e^{-\alpha t} e^{-j\omega t} dt = \frac{1}{\alpha + j\omega}$$

$$|F(\omega)| = \frac{1}{\sqrt{\alpha^2 + \omega^2}}$$



$$\varphi(\omega) = -\arctan\left(\frac{\omega}{\alpha}\right)$$



The inverse Fourier transform of the spectrum function $X(\omega) = \frac{1}{j\omega + 2}$ is ().

☒ A $e^{-2t}u(t)$

☐ B $te^{-2t}u(t)$

☐ C $e^{-0.5t}u(t)$

☐ D $te^{2t}u(t)$

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2. Odd-symmetric Two-sided Exponential Signal

$$f(t) = e^{-\alpha t} u(t) - e^{\alpha t} u(-t)$$

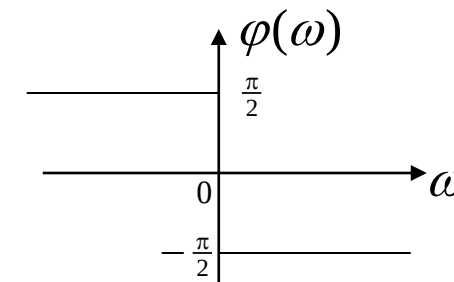
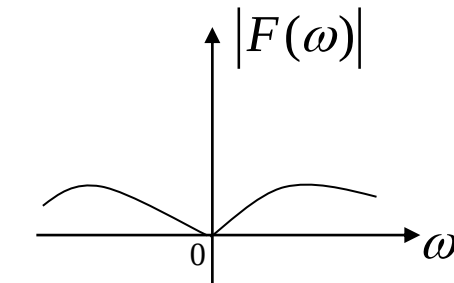
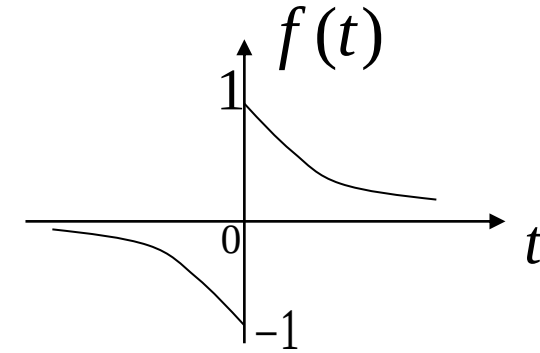
$$= f_1(t) + f_2(t)$$

$$F_1(\omega) = \frac{1}{\alpha + j\omega}$$

$$F_2(\omega) = \int_{-\infty}^{\infty} f_2(t) e^{-j\omega t} dt = \int_{-\infty}^0 -e^{\alpha t} e^{-j\omega t} dt = \frac{-1}{\alpha - j\omega}$$

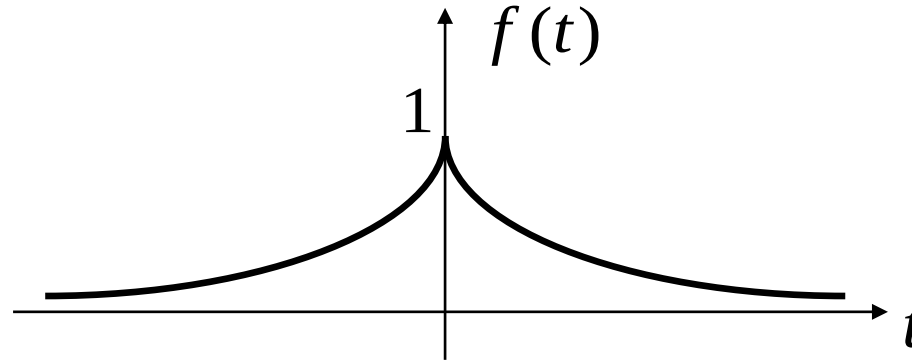
$$F(\omega) = F_1(\omega) + F_2(\omega) = \frac{-2j\omega}{\alpha^2 + \omega^2}$$

$$|F(\omega)| = \frac{2|\omega|}{\alpha^2 + \omega^2}, \quad \varphi(\omega) = \begin{cases} \pi/2 & \omega < 0 \\ -\pi/2 & \omega > 0 \end{cases}$$

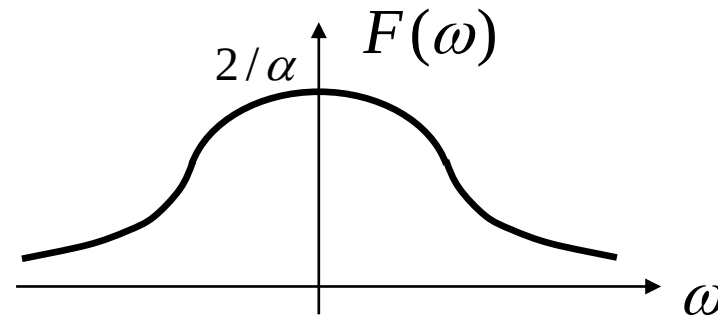


3. Even-symmetric Two-sided Exponential Signal

$$f(t) = e^{-\alpha|t|} \quad (\alpha > 0)$$



$$F(\omega) = \frac{2\alpha}{\alpha^2 + \omega^2}$$

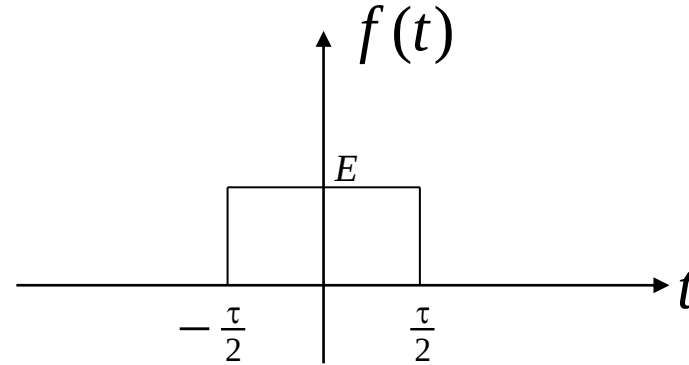


$$|F(\omega)| = \frac{2\alpha}{\alpha^2 + \omega^2}$$

$$\phi(\omega) = 0$$

4. Rectangular Pulse Signal (Gate Function)

$$f(t) = E[u(t + \frac{\tau}{2}) - u(t - \frac{\tau}{2})]$$

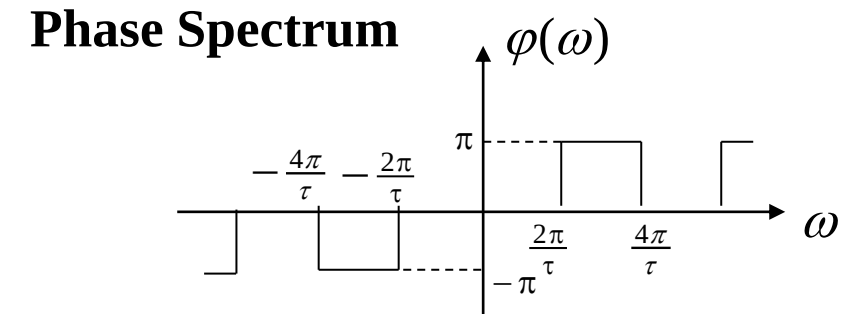
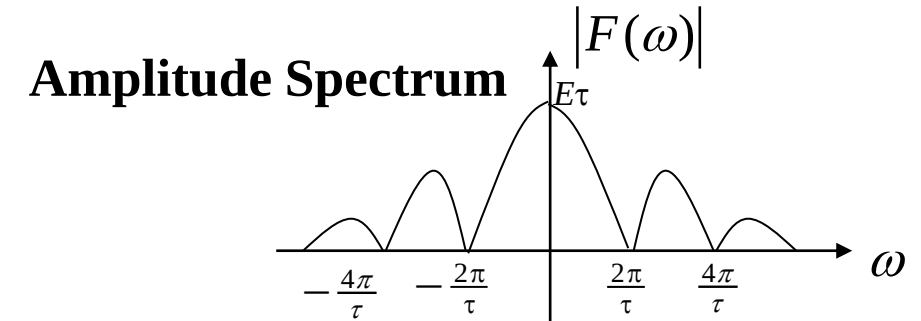
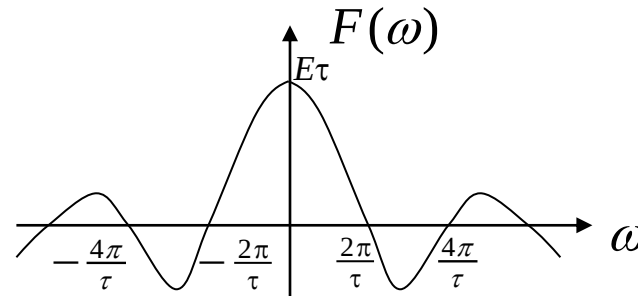
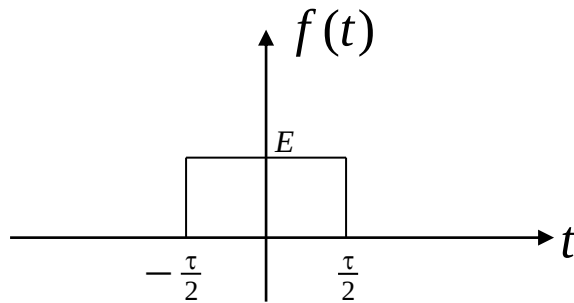


$$\begin{aligned} F(\omega) &= \int_{-\infty}^{\infty} f(t)e^{-j\omega t} dt = \int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} Ee^{-j\omega t} dt = E \frac{1}{-j\omega} e^{-j\omega t} \Big|_{-\frac{\tau}{2}}^{\frac{\tau}{2}} \\ &= E \frac{e^{-j\omega \frac{\tau}{2}} - e^{j\omega \frac{\tau}{2}}}{-j\omega} = 2E \frac{\sin\left(\frac{\tau\omega}{2}\right)}{\omega} = E\tau \text{Sa}\left(\frac{\tau\omega}{2}\right) \end{aligned}$$

3.5 Fourier Transform of Typical Aperiodic Signals



$$E[u(t + \frac{\tau}{2}) - u(t - \frac{\tau}{2})] \xleftrightarrow{FT} E\tau \text{Sa}(\omega \frac{\tau}{2})$$



Equivalent Bandwidth

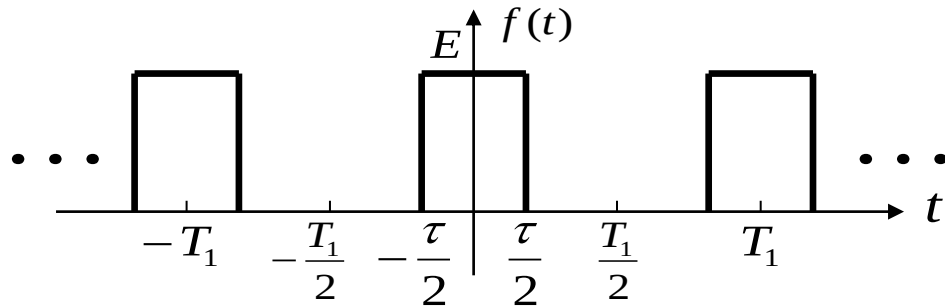
It can be seen from the above spectrum diagram of the rectangular wave that the signal energy is mainly concentrated within the frequency range $0 \sim 2\pi / \tau$. This range, $B\omega$ is called the **equivalent bandwidth** of the rectangular wave.

$$B_{\omega} = \frac{2\pi}{\tau} \text{ (rad/s) or } \frac{1}{\tau} \text{ (Hz)}$$

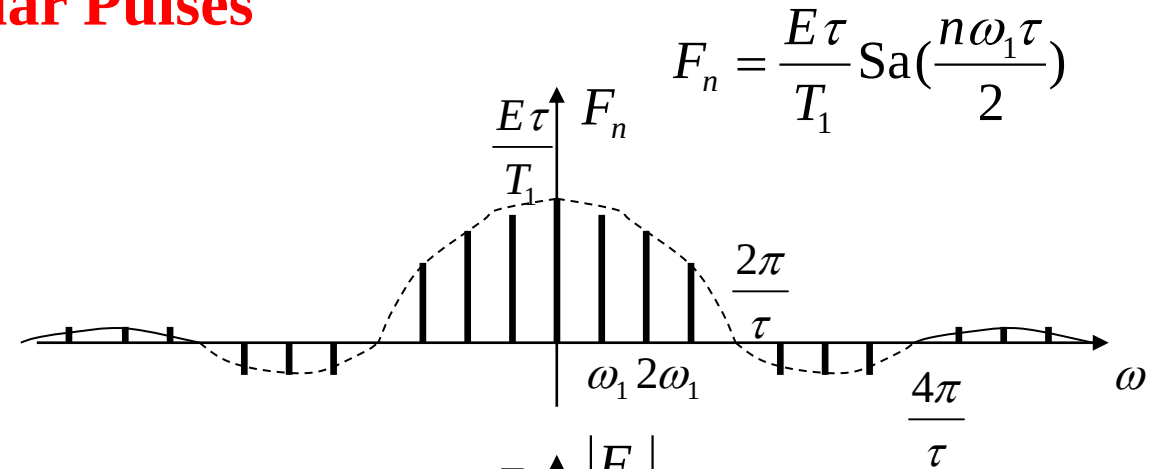
3.5 Fourier Transform of Typical Aperiodic Signals



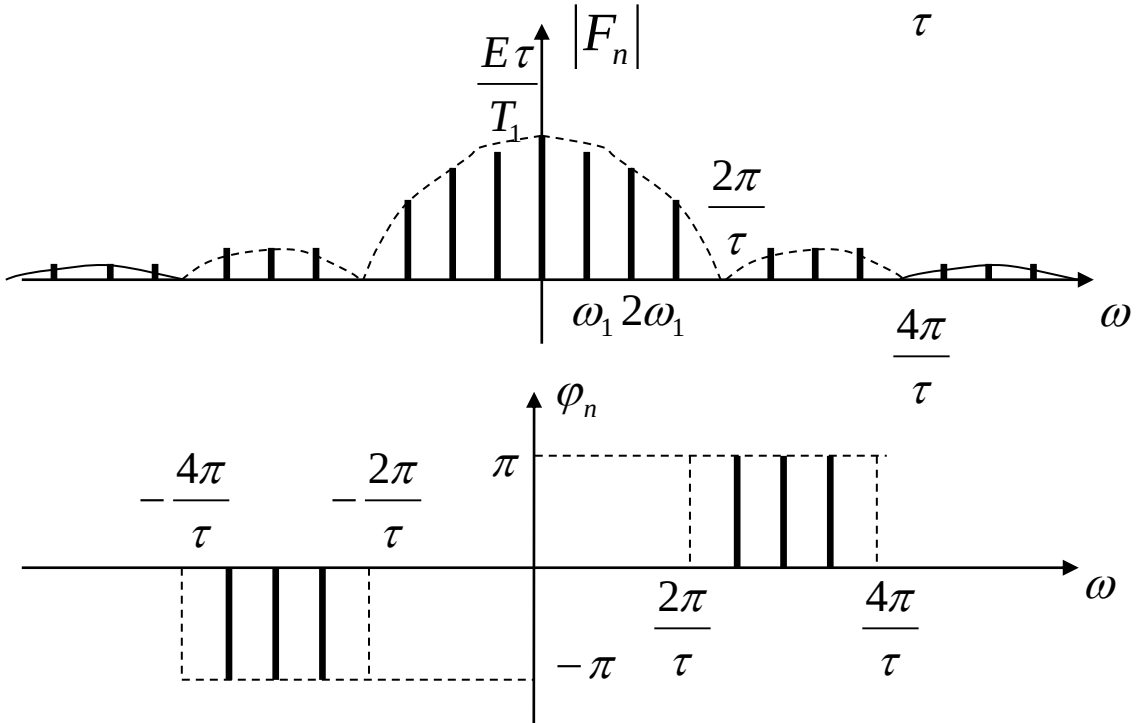
Review: Spectrum of Periodic Rectangular Pulses



Amplitude
Spectrum



Phase
Spectrum



3.5 Fourier Transform of Typical Aperiodic Signals



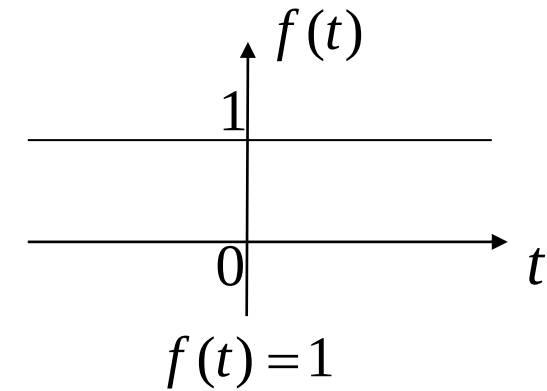
5. DC Signal

A DC signal is a constant (set to 1 here).

It does not satisfy the absolutely integrable condition, so its Fourier transform cannot be directly obtained by integration—we use the **impulse function $\delta(\omega)$** to derive it.

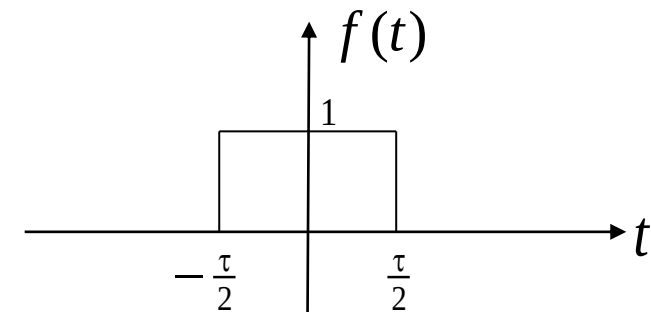
This result can be obtained by taking the limit of τ to ∞ for a rectangular pulse with amplitude 1 and width τ :

$$f(t) = 1 = \lim_{\tau \rightarrow \infty} \left[u\left(t + \frac{\tau}{2}\right) - u\left(t - \frac{\tau}{2}\right) \right]$$

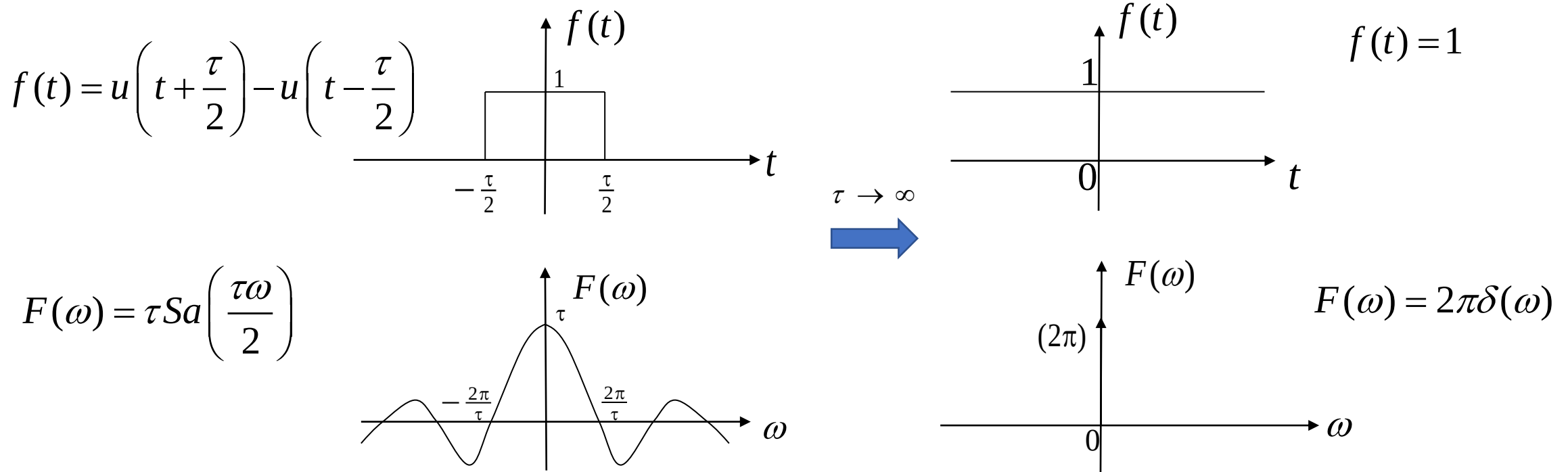


$\tau \rightarrow \infty$ 

$$f(t) = u\left(t + \frac{\tau}{2}\right) - u\left(t - \frac{\tau}{2}\right)$$



3.5 Fourier Transform of Typical Aperiodic Signals



When $\tau \rightarrow \infty$, the spectrum width tends to 0, and the amplitude tends to infinity (forming an **impulse function**), the area of $F(\omega)$ (strength of the impulse function) is:

$$\int_{-\infty}^{\infty} \tau \text{Sa}\left(\frac{\tau\omega}{2}\right) d\omega = 2 \int_{-\infty}^{\infty} \text{Sa}\left(\frac{\tau\omega}{2}\right) d\left(\frac{\tau\omega}{2}\right) = 2\pi \quad \left(\int_{-\infty}^{\infty} \text{Sa}(x) dx = \pi \right)$$

So

$$1 \xleftrightarrow{FT} 2\pi\delta(\omega)$$

$$\int_{-\infty}^{\infty} e^{-j\omega t} dt = 2\pi\delta(\omega)$$

6. Signum Function Signal

$$f(t) = \text{sgn}(t)$$

The signum function **does not satisfy the absolutely integrable condition.**

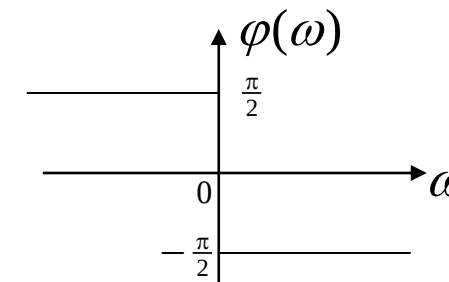
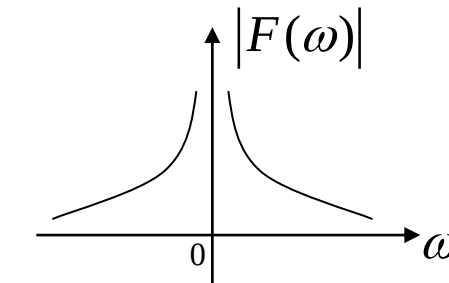
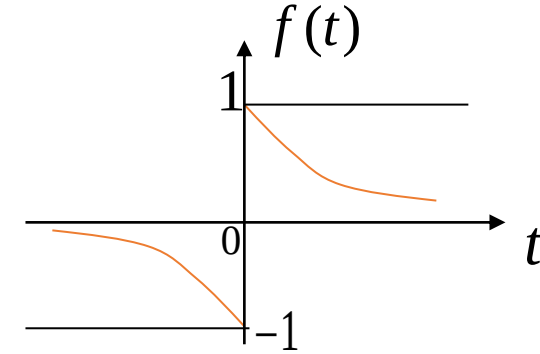
Its Fourier transform is derived by taking $\alpha \rightarrow 0$ for the odd-symmetric two-sided exponential signal:

$$f(t) = \text{sgn}(t) = \lim_{\alpha \rightarrow 0} [-e^{\alpha t} u(-t) + e^{-\alpha t} u(t)]$$

So

$$F(\omega) = \lim_{\alpha \rightarrow 0} \frac{-2j\omega}{\alpha^2 + \omega^2} = -j \frac{2}{\omega} = \frac{2}{j\omega}$$

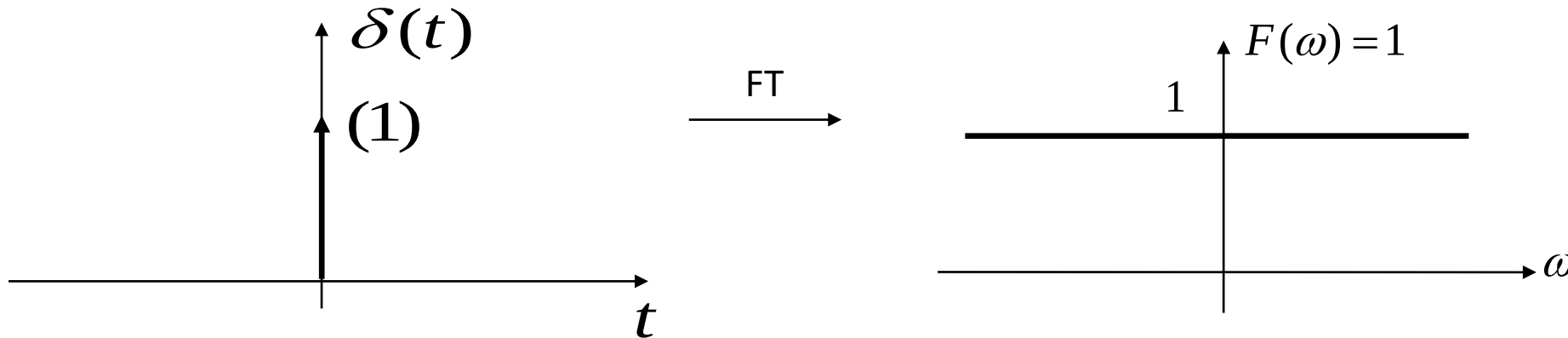
$$|F(\omega)| = \frac{2}{|\omega|}, \quad \varphi(\omega) = \begin{cases} \pi/2 & \omega < 0 \\ -\pi/2 & \omega > 0 \end{cases}$$



1. Fourier Transform of Impulse Function

$$F(\omega) = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} \delta(t) dt = 1$$

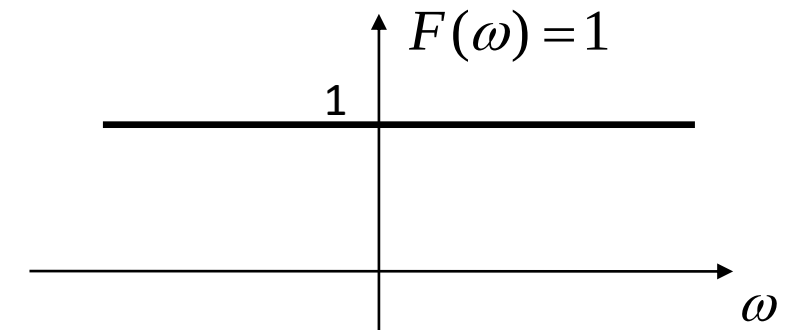
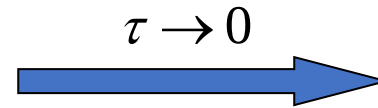
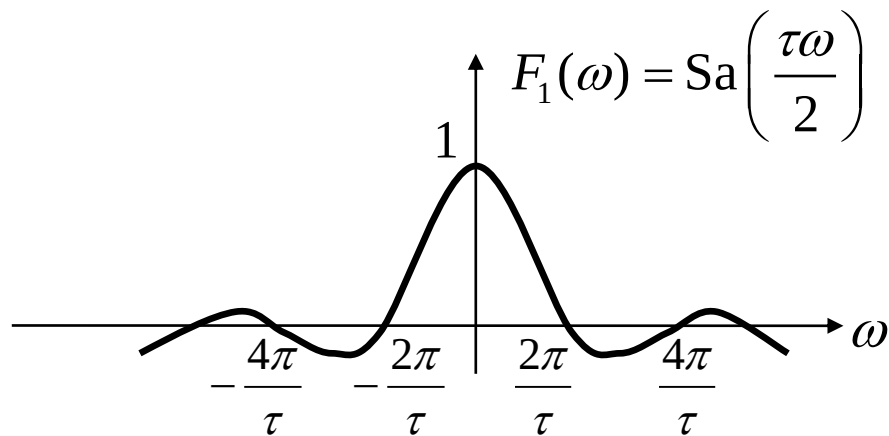
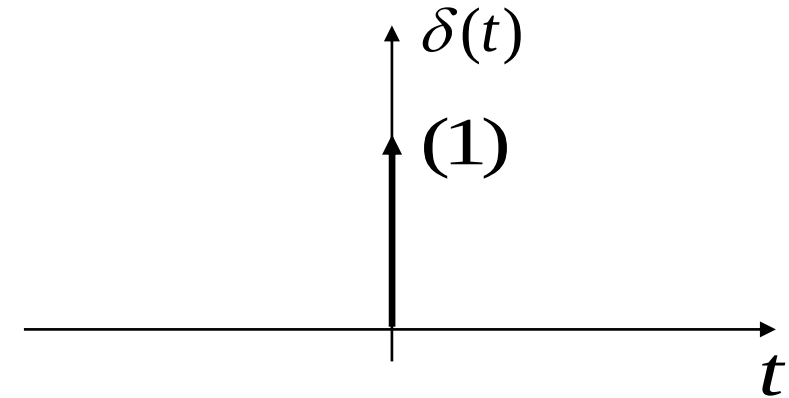
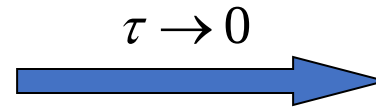
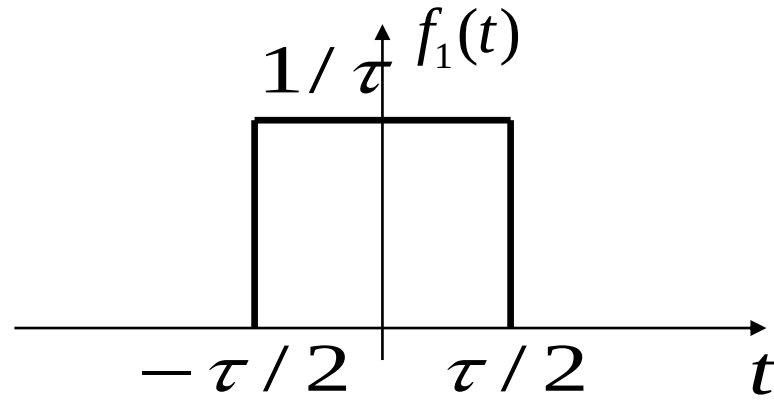
The spectrum of the unit impulse function is a constant, meaning the spectrum is uniform across the entire frequency range. This type of spectrum is often called a "**uniform spectrum**" or "**white spectrum**".



3.6 Fourier Transform of Impulse and Step Functions



Deriving Impulse Function's Fourier Transform from Rectangular Pulse



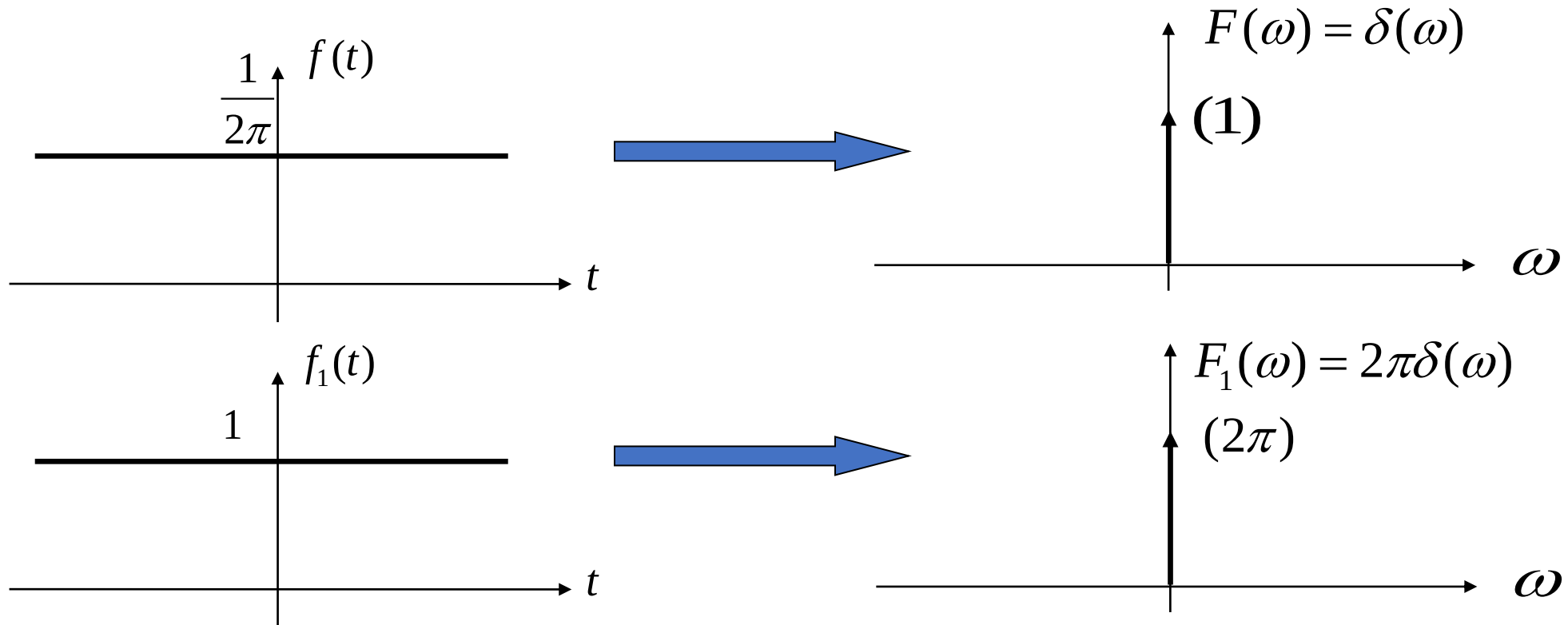
3.6 Fourier Transform of Impulse and Step Functions



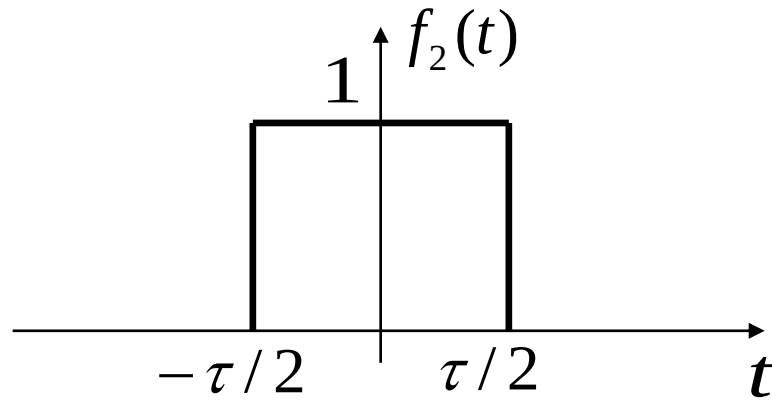
2. Inverse Fourier Transform of Impulse Function

$$f(t) = F^{-1}[\delta(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega) e^{j\omega t} d\omega = \frac{1}{2\pi}$$

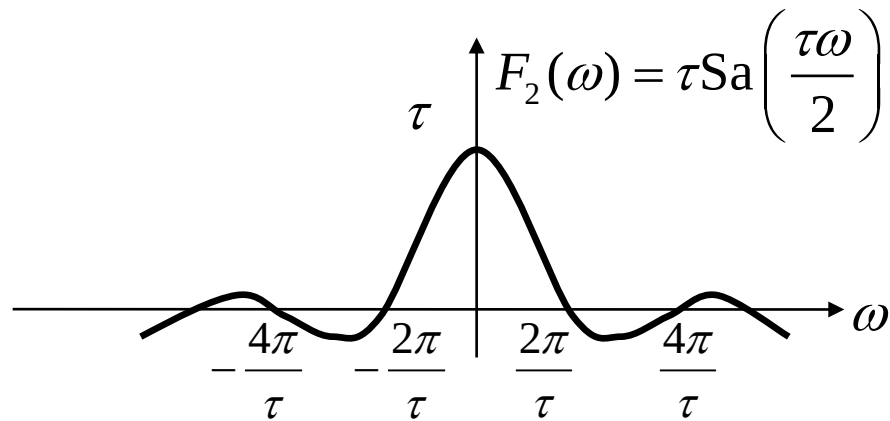
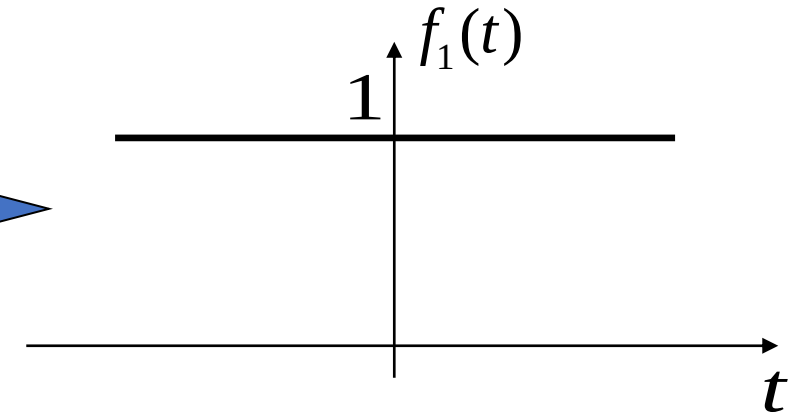
$$\text{或 } F\left[\frac{1}{2\pi}\right] = \delta(\omega), \quad F[1] = 2\pi\delta(\omega)$$



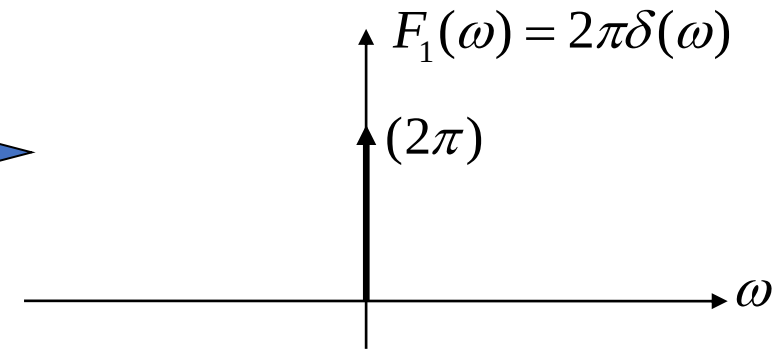
3.6 Fourier Transform of Impulse and Step Functions



$\tau \rightarrow \infty$



$\tau \rightarrow \infty$



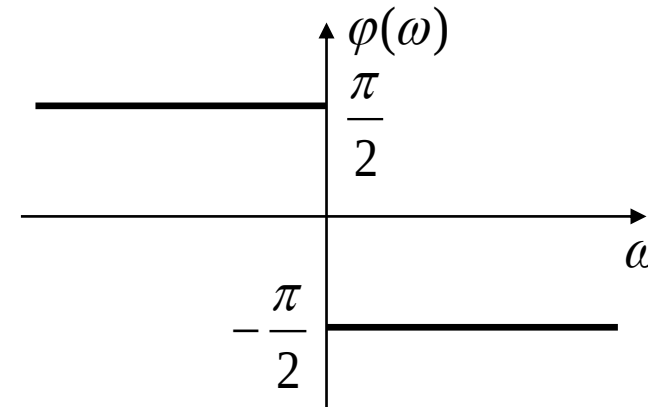
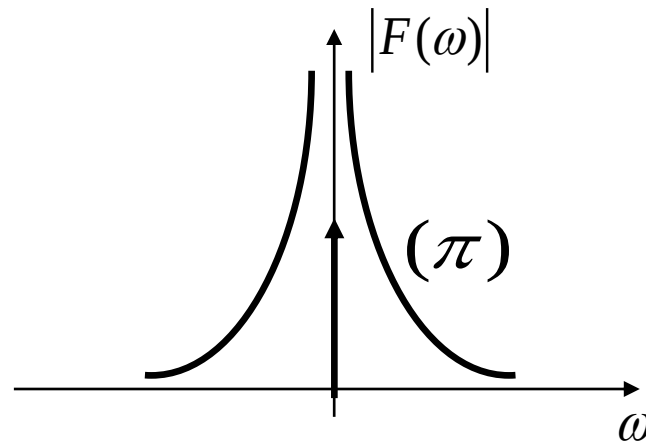
3. Fourier Transform of Step Function

$$\because u(t) = \frac{1}{2} + \frac{1}{2} \text{sgn}(t)$$

$$\begin{aligned} \therefore F(\omega) &= \mathcal{F}[u(t)] = \mathcal{F}\left[\frac{1}{2}\right] + \mathcal{F}\left[\frac{1}{2} \text{sgn}(t)\right] \\ &= \pi\delta(\omega) + \frac{1}{j\omega} \end{aligned}$$

corresponding to
DC component

other frequency components caused by
the jump at $t=0$



- Repeat the examples in the lecture notes.
- 3.54(b) and 3.55(a)(b) in the textbook.